DATA SCIENCE & ARTIFICIAL INTELLIGENCE & CS/IT

Calculus and Optimization

Limit, Continuity and differentiablity

Lecture No. 01



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

FUNCTIONS & THEIR GRAPHS

Topics to be Covered



Topic

BASICS of LIMITS & CONTINUITY

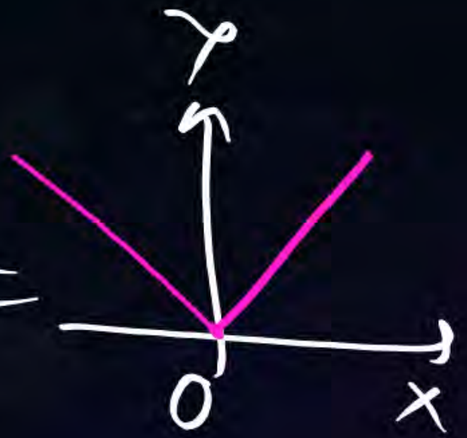
(Part-1)

RECAP

PIECEWISE func^n → If func^n is defined by Multiple sub function s.t, Domain of each subfunction is different then function is called Piecewise func^n

for eg. Mod func^n, Signum func^n, G.I.F, L.I.F, Fractional Part func^n etc

MOD func^n: $y = |x| = \sqrt{x^2} = \begin{cases} -x, & x < 0 \\ +x, & x \geq 0 \end{cases} = \max(x, -x) =$



REAL

Signum Function → it is defined as;

$$y = f(x) = Sgn(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} = \begin{cases} -1, & x < 0 \\ +1, & x > 0 \\ 0, & x = 0 \end{cases}$$



Domain = $(-\infty, \infty)$

Range = $\{-1, 0, 1\}$ is Range of Sgn Func^n Consisting only 3 Points $+1, -1$ & 0

& it is an ODD Func^n.

eg $Sgn(0.4) = 1$, $Sgn(4.9) = 1$, $Sgn(-1.2) = -1$, $Sgn(0) = 0$ .....

G.I.F: $y = f(x) = [x] = ?$

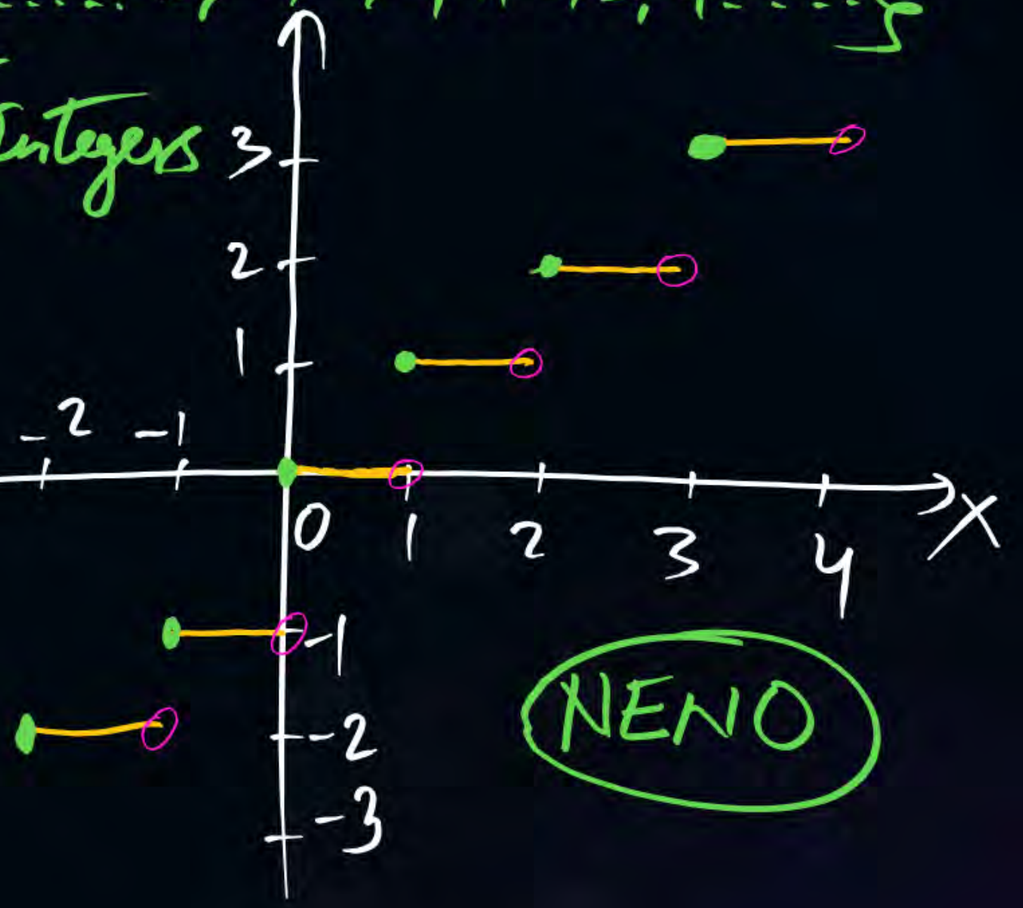
(floor func^n)

Domain $= (-\infty, \infty)$

Range $= \{\ldots -3, -2, -1, 0, 1, 2, 3, 4 \ldots\}$

$=$ Set of Integers

$$y = [x] = \begin{cases} \vdots & \\ -2 & ; \ -2 \le x < -1 \\ -1 & ; \ -1 \le x < 0 \\ 0 & ; \ 0 \le x < 1 \\ 1 & ; \ 1 \le x < 2 \\ 2 & ; \ 2 \le x < 3 \\ 3 & ; \ 3 \le x < 4 \\ \vdots & \end{cases}$$



NENO

Ref^{n} of Ceiling Function $y=\lceil x \rceil = ?$

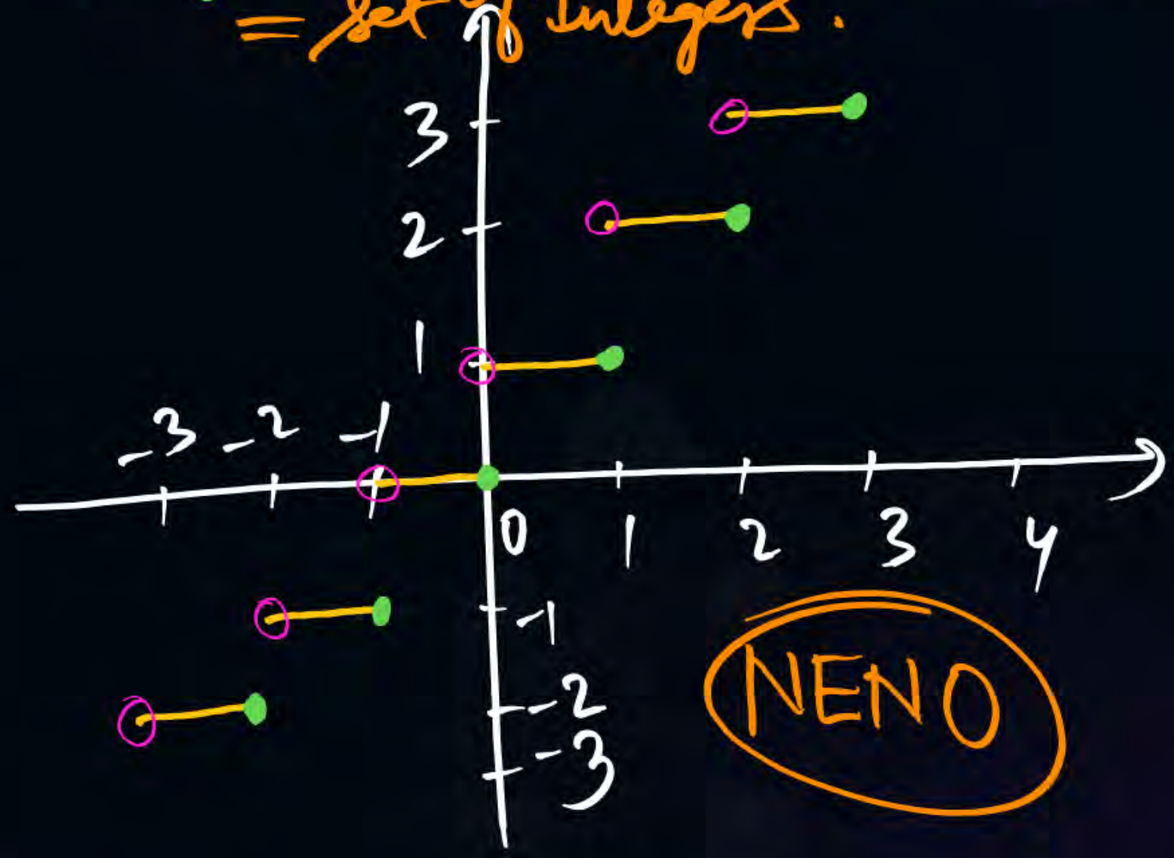
(L.I.F)

Domain $= (-\infty, \infty)$

Range $= \{\ldots -2, -1, 0, 1, 2, 3 \ldots\}$

$=$ set of Integers.

$$y=\lceil x \rceil = \begin{cases} \vdots & \vdots \\ -1 & , -2 < x \le -1 \\ 0 & , -1 < x \le 0 \\ \mathbf{1} & , 0 < x \le 1 \\ 2 & , 1 < x \le 2 \\ 3 & , 2 < x \le 3 \\ 4 & , 3 < x \le 4 \\ \vdots & \vdots \end{cases}$$



NENO

Fractional Part func^n –

$$y = f(x) = \{x\} = x - \lfloor x \rfloor$$

eg $\{0.4\} = 0.4 - 0 = 0.4$

$\{0.9\} = 0.9 - 0 = 0.9$

$\{1.2\} = 1.2 - 1 = 0.2$

$\{-1.2\} = -1.2 - (-2) = 0.8$

$\{2.9\} = 2.9 - (2) = 0.9$

$\{-2.9\} = -2.9 - (-3) = 0.1$

- - - - - - -

$\lfloor -5.3 \rfloor = -6$

$\lceil -5.3 \rceil = -5$

$\{-5.3\} = -5.3 - (-6) = 0.7$

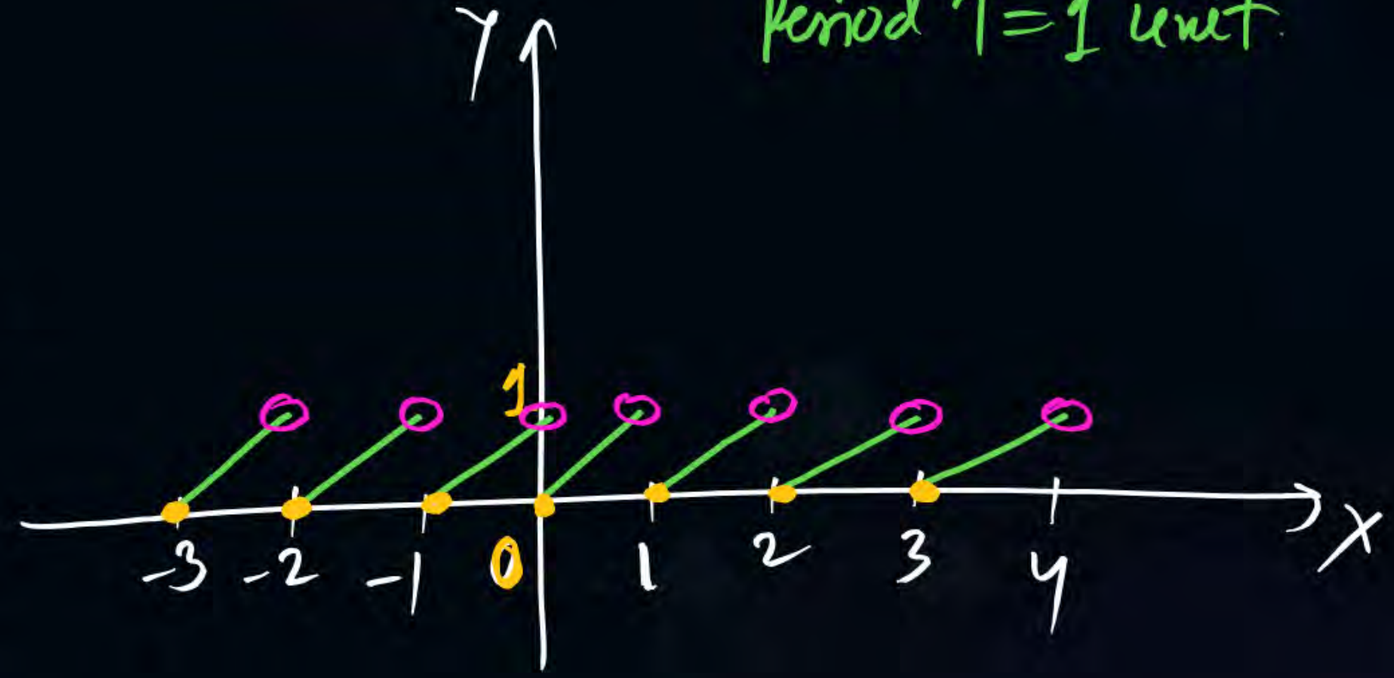
$\{3\} = 0, \{-3\} = 0$

Defⁿ: $y = \{x\} = x - \lfloor x \rfloor = ?$

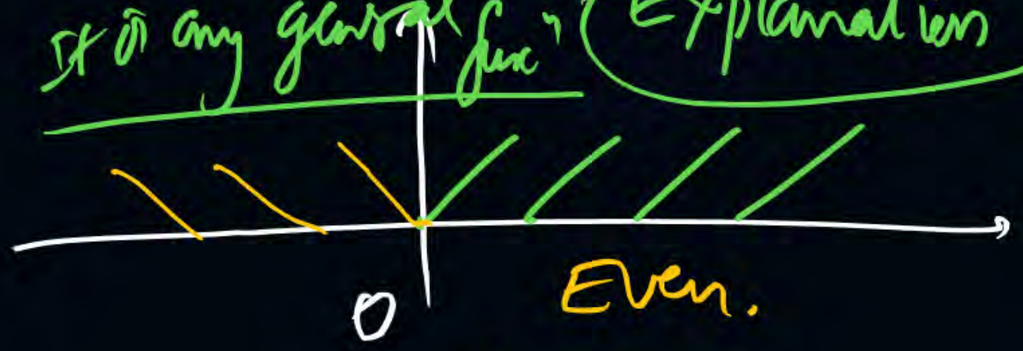
Domain $= (-\infty, \infty)$, MENO,

Range $= [0, 1)$, PERIODIC with Period $T = 1$ unit

$$y = \{x\} = \begin{cases} x+2, & -2 \le x < -1 \\ x+1, & -1 \le x < 0 \\ x, & 0 \le x < 1 \\ x-1, & 1 \le x < 2 \\ x-2, & 2 \le x < 3 \\ x-3, & 3 \le x < 4 \\ \vdots & \end{cases}$$



If f is any general func" (Explanations)





F.P.f

$\{0.9\} = 0.9$, $\{1.2\} = 0.2$

$\{0.1\} = 0.1$, $\{1.5\} = 0.5$

$\{0\} = 0$, $\{1\} = 0$

$\{2\} = 0$

$\{2.0\} = 0$

$\{2.1\} = 0.1$

$\{2.99\} = 0.99$

is Range $\{x\} = [0, 1)$

Domain: Set of permissible values of x is called Domain of $y=f(x)$

& there is **No Shortcut Method** to find Domain of given func^n, knowledge. it can be Calculated only by using Common sense or by previous

eg: Find the Domain of following func^n;

(1) $y=f(x)=\dfrac{1}{x^2-5x+6}$

$$f(x)=\frac{1}{(x-2)(x-3)}$$

Domain: $R-\{2,3\}$

(2) $y=\sqrt{-x^2+5x-6}$

for the validity of above func^n, we must have,

$$-x^2+5x-6 \geq 0$$
$$x^2-5x+6 \leq 0$$
$$(x-2)(x-3) \leq 0$$
$$2 \leq x \leq 3$$

i.e. Domain $=[2,3]$

eg $y=\sqrt{-x^2+5x-6}$, Dom$=[2,3]$

Let us take $x=1$ then

$y=\sqrt{-1^2+5(1)-6}=\sqrt{-2}=$ Not Real

Let us take $x=5$ then

$y=\sqrt{-5^2+5(5)-6}=\sqrt{-6}=$ Not Real.

ie Permissible values of x lies only in b/w 2 & 3

eg let $x=2.5$

$y=\sqrt{-(2.5)^2+5(2.5)-6}$

$=\sqrt{-6.25+12.5-6}$

$=\sqrt{-12.25+12.50}$

$=\sqrt{0.25}=0.5$

ie y is also Real.

ie Valid

Common Sense:

① $\frac{1}{f(x)}$; $f(x) \neq 0$

② $\sqrt{f(x)}$; $f(x) \geq 0$

③ $\frac{1}{\sqrt{f(x)}}$; $f(x) > 0$

④ $\log_e(f(x))$; $f(x) > 0$

⑤ $\sin(f(x))$; $-\infty < f(x) < \infty$

Q: $y = f(x) = \log(x-3)$,

W.K. that for $\log f(x)$, $f(x) > 0$

So In this question $x-3 > 0 \Rightarrow x > 3$

is Domain $= (3, \infty)$

Q: $y = f(x) = \sin(x-3)$

W.K. that for $\sin f(x)$; $-\infty < f(x) < \infty$

$-\infty < (x-3) < \infty$

$-\infty < x < \infty$

is Domain $= (-\infty, \infty)$

Q $y = \sin^{-1}(x-3)$

Sol: Here $\sin y = x-3$

W.K.T,

$$-1 \le \sin y \le 1$$

$$-1 \le x-3 \le 1$$

$$-1+3 \le x \le 1+3$$

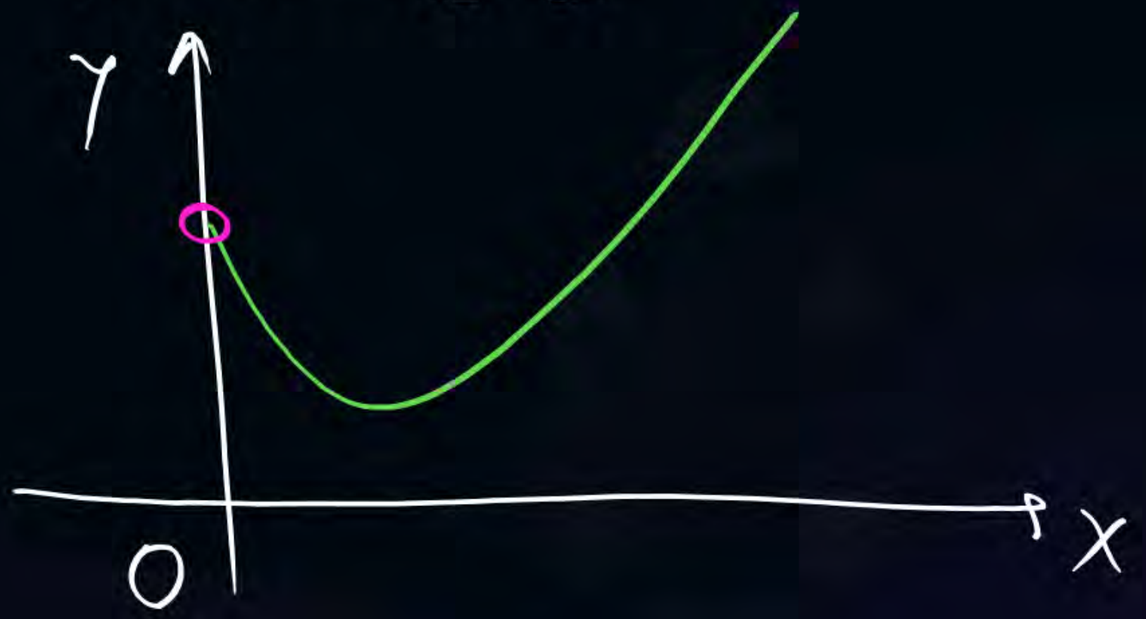
$$2 \le x \le 4$$

i.e Dom $= [2,4]$

Q $y = x^x = e^{\log_e(x^x)} = e^{x \log x}$

$\because y = x^x$ contains $\log x$ also so

Domain $= (0, \infty)$



Analysis (PODCAST) → for $y = x^x$; Domain = $(0, \infty)$

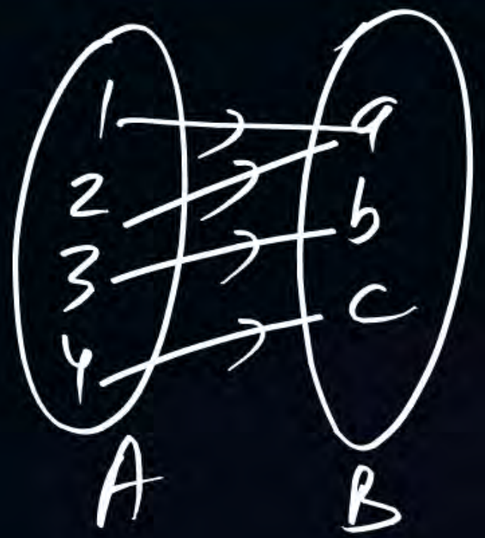
eg: Evaluate $(-2)^{-2} = ? = \frac{1}{(-2)^2} = \frac{1}{4}$ ✓

eg if $y = x^x$ then evaluate $y(-2) = ?$ = DNE

Dom : $(0, \infty)$

Function: func^n is a special type of Relationship under certain Restrictions & these Restrictions are called Domain of the func^n.

Some More Concepts :

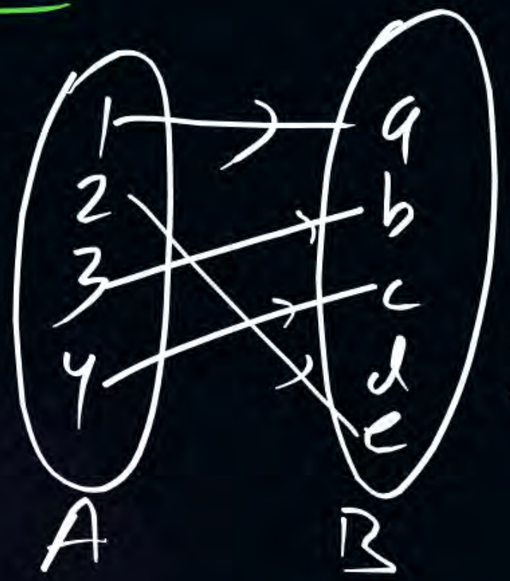


Many one / ONTO

Range = {a, b, c}

& Range = Codomain

So ONTO



one one / INTO

Range = {a, b, c, e}

∴ Range ⊂ Codomain

Hence INTO



Many one / INTO

∴ Range = {b, c}

∴ Range ⊂ Codomain

is INTO

Constant fnc :



MANY ONE / ONTO.

Note ① if Different elements have different Images then f is one-one

i.e $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$ $\forall x_1, x_2$

② if Different elements have same Images then f is MANY ONE

i.e $x_1 \neq x_2 \Rightarrow f(x_1) = f(x_2)$

Shortcut : if Any Random line $\parallel^{el}$ to X axis Cuts the graph only at one point then f is Called one-one otherwise Many one.

Eg Let $y = f(x) = x^2$



i.e $y = x^2$ is Many one func^n.

① one-one Mapping ≃ INJECTIVE Mapping

② one-one / ONTO Mapping ≃ BIJECTIVE Mapping

one-one Correspondence

If $y = f(x) = x^2 ; R \to R$ then find Range = ? $= [0, \infty)$

Domain, Codomain ∵ Range ⊂ Codomain ⇒ INTO

If $y = f(x) = x^2 : R \to R^+$ then find it's Nature?

Dom, Codom. & Range $= R^+$ Hence ONTO

BASICS of LIMITS

Infinity → Infinity is Not a very large Number. It is the presentation of that Concept which is beyond Imagination.

→ & Infinity is Not unique, it depends upon the Imagination capacity of an Individual.

→ $\frac{\text{Something}}{\infty} \cong 0$ (Assumption)

$\frac{\text{Something}}{0} = \text{N.D.}$ (FACT)

GOD GOD GOD

→ $\infty + \infty = \infty$ ≈ N.D

I don't know — you don't know — we don't know

→ $\infty \times \infty = \infty$ ≈ N.D

→ $\infty^{\infty} = \infty$ ≈ N.D

→ $\infty - \infty = ?$ = IND form.

→ $\frac{\infty}{\infty} = ?$ = IND form.

→ $\infty^{0} = ?$ = IND form.

→ $1^{\infty} = ?$ = IND form.

→ $0 \times \infty = ?$ = IND form.

→ $0^{0} = ?$ = IND form.

→ $\frac{0}{0} = ?$ = IND form.

INDETERMINATE form → There are exactly 7 Indeterminate forms for eg $\frac{0}{0}, \frac{\infty}{\infty}, 0\times\infty, \infty-\infty, 0^0, \infty^0, 1^\infty$

PROPERTIES of LOG (on Students Demand)

See Next Slide

Def^n: if $\boxed{\log_a x = b \iff a^b = x}$ $\because \log_4(64) = 3 \Leftrightarrow 4^3 = 64$

Here a is taken as the number i.e. $(a>0)$, $\Rightarrow a^b > 0$ i.e. $(x>0)$ But b may be $-ve$.

Note → ① $\boxed{\log_a a = 1}$, $a>0$ & $a \neq 1$

eg $\log_7 7 = 1$, $\log_5 5 = 1$, $\log_1 1 = \text{N.D}$

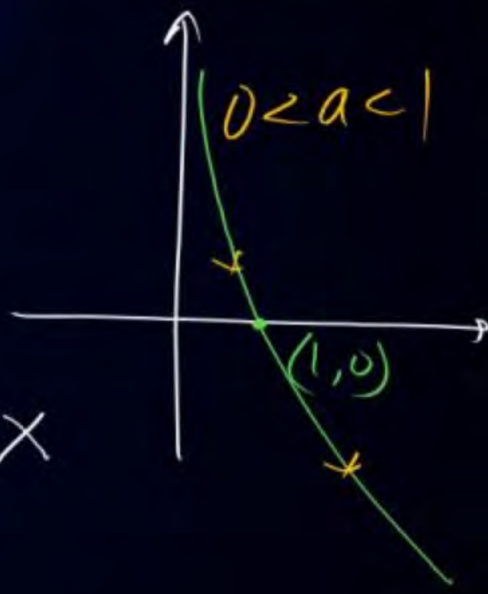
② $\boxed{\log_a 1 = 0}$,

③ $\boxed{\log_a(\infty) = \infty}$ (for $a>1$)

④ $\boxed{\log_a(0) = -\infty}$ (for $a>1$) $\left(\because a^{-\infty} = \frac{1}{a^{\infty}} = \frac{1}{\infty} = 0\right)$

⑤ Graph of $(y = \log_a x)$.





⑥ $\log_a x = \log_a y \iff x = y$

Proof: Let $\log_a x = \alpha \Rightarrow a^\alpha = x$

& $\log_a y = \alpha \Rightarrow a^\alpha = y$

$\Rightarrow x = y$

⑦ if $a > 1$, $\log_a x > \log_a y \iff x > y$

if $0 < a < 1$, $\log_a x > \log_a y \iff x < y$

⑧ Product formula

$$\log_a(mn) = \log_a m + \log_a n$$

Let $\log_a m = \alpha \Rightarrow a^\alpha = m$

& $\log_a n = \beta \Rightarrow a^\beta = n$

Now $a^{\alpha+\beta} = a^\alpha \cdot a^\beta = m \cdot n$

$\because$ when $2^3 = 8 \Rightarrow \log_2 8 = 3$

$\log_a(mn) = \alpha + \beta = \log_a m + \log_a n$

(9) Quotient Formula –

$$\log_a\left(\frac{m}{n}\right) = \log_a(m) - \log_a(n)$$

Proof: Let $\log_a(m) = \alpha \Longrightarrow a^{\alpha} = m$

& $\log_a(n) = \beta \Longrightarrow a^{\beta} = n$

Now $a^{\alpha-\beta} = a^{\alpha} \cdot a^{-\beta} = \frac{a^{\alpha}}{a^{\beta}} = \frac{m}{n}$

$\because 2^3 = 8 \Longrightarrow \log_2 8 = 3$

$$\log_a\left(\frac{m}{n}\right) = \alpha - \beta = \log_a m - \log_a n$$

(10) Power formula –

$$\log_a(m^n) = n \cdot \log_a m$$

Proof: $\log_a(m^n) = \log_a(\underbrace{m \cdot m \cdot m \cdots m}_{n \text{ times}})$

$= \underbrace{\log_a m + \log_a m + \log_a m + \cdots + \log_a m}_{n \text{ times}}$

$= n \log_a m$

(11) Base change formula →

$$\log_y(x) = \frac{\log_a x}{\log_a y}$$

Let $\log_y(x) = \alpha \implies y^\alpha = x$

Now $\log_a(x) = \log_a(y^\alpha) = \alpha \log_a(y)$

$$\log_a x = \log_y x \cdot (\log_a y)$$

i.e. $$\log_y x = \frac{\log_a x}{\log_a y}$$ //

(*) Why $\log_1 x = \text{N.D}$?

Reason: $$\log_1 x = \frac{\log_a x}{\log_a 1} = \frac{\log_a x}{0} = \text{ND}$$

(*) Why $\log_1 1 = \text{ND}$?

Reason: $$\log_1 1 = \frac{\log_a 1}{\log_a 1} = \frac{0}{0} = \text{IND form}$$

(12) Reciprocal formula →

$$\log_y x = \frac{\log_x x}{\log_x y} = \frac{1}{\log_x y}$$

$$\log_y x = \frac{1}{\log_x y}$$

(13) $\log_{a^b}(x) = \frac{1}{b}\log_a x$

LHS $= \log_{a^b}(x) = \frac{\log_a(x)}{\log_a(a^b)} = \frac{\log_a x}{b \log_a a}$

$= \frac{1}{b}\log_a x = RHS$

(14) $a = e^{\log_e a}$ if $a > 0$

Let $e^{\log_e a} = \alpha \Rightarrow \log_e \alpha = \log_e a \Rightarrow \alpha = a$

$\because \left(2^3 = 8 \Rightarrow \log_2 8 = 3\right)$

i.e. $e^{\log_e a} = a$

(15) $a^x = e^{x \log_e a}$; $a > 0$

W.K.T. that, $a = e^{\log_e a}$

$\Rightarrow a^x = e^{\log(a^x)} = e^{x \log_e a}$ Hence Proved

(16) $a^{\log_e b} = b^{\log_e a}$

$a^{\log_e b} = \left(e^{\log_e a}\right)^{\log_e b} = e^{\log a \cdot \log b}$

$= \left(e^{\log_e b}\right)^{\log_e a} = e^{\log_e b \cdot \log_e a}$

$= b^{\log_e a}$

$=$

THANK - YOU

Tel: drpuneet Sir pw